

CS 170 Proficiency Quiz 3

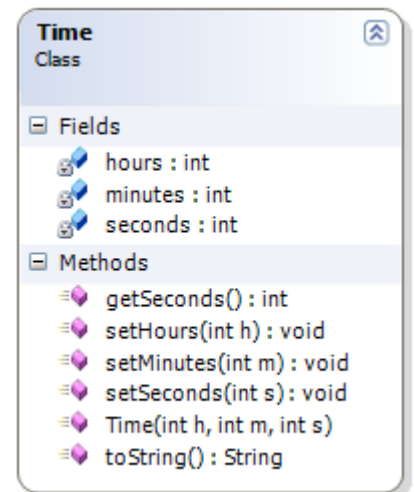
Write a program that **defines the class described below**. Carefully read the class specification; while it is similar to the classes you have used in Chapter 3, **it is not the same**. You will have 30 minutes to complete the problem. You may not access any notes or files, other than the Java quick reference I've supplied, or communicate with anyone in any way. When you have finished and submitted the quiz, check with me to be sure I have received it.

You can check your problem by using **Tools->Compile** to see if your code is syntactically correct. To test your code, use **Tools->Go**. Since this is a class, and not a program, you cannot run your code to see what it looks like.

Defining Classes

Write and document the definition for the **Time** class. The **Time** class represents time using the 24-hour (military) time format.

- ✓ Each **Time** object has three instance variables
- ✓ The class has one constructor that takes three arguments in the order **hours**, **minutes** and **seconds**.
- ✓ The class has one accessor method named **getSeconds()** which returns the **total number of seconds** that the **Time** object contains. (Multiply **hours** by **3600** and **minutes** by **60**; add those two sums to **seconds** and return that amount.)
- ✓ The class has three mutator methods—**setHours()**, **setMinutes()** and **setSeconds()**—that modify the corresponding instance variables.
- ✓ The class has a **toString()** method that returns the time as a String in the form: **23:15:3**. Use concatenation. You do not have to format the time



Grading: 80% correctness, 20% style